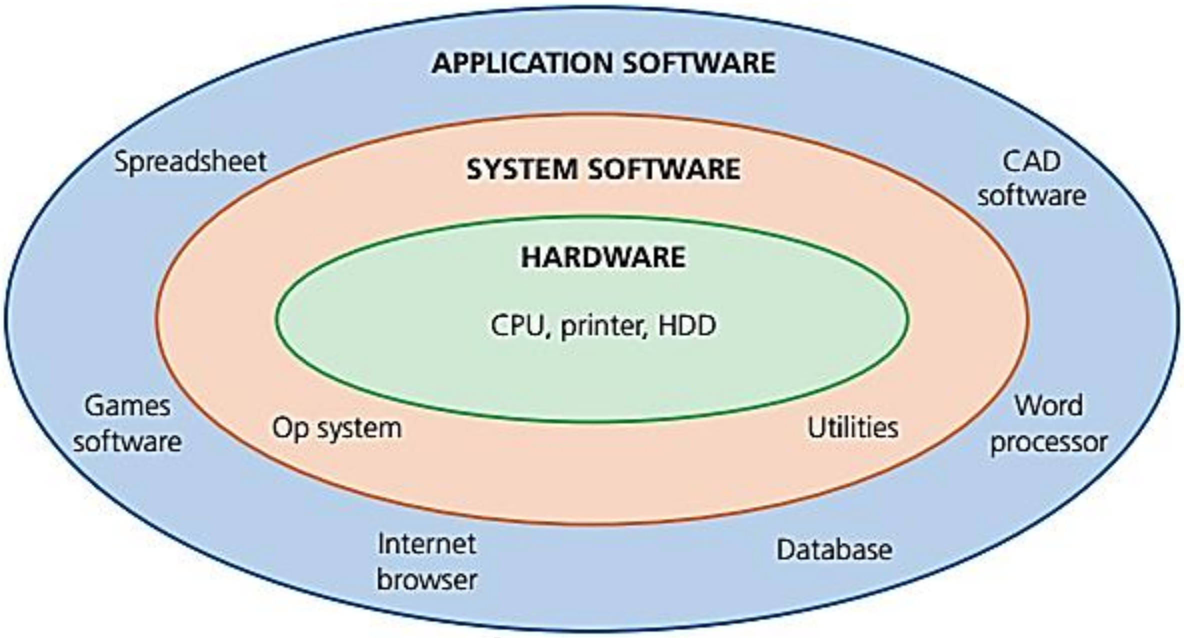
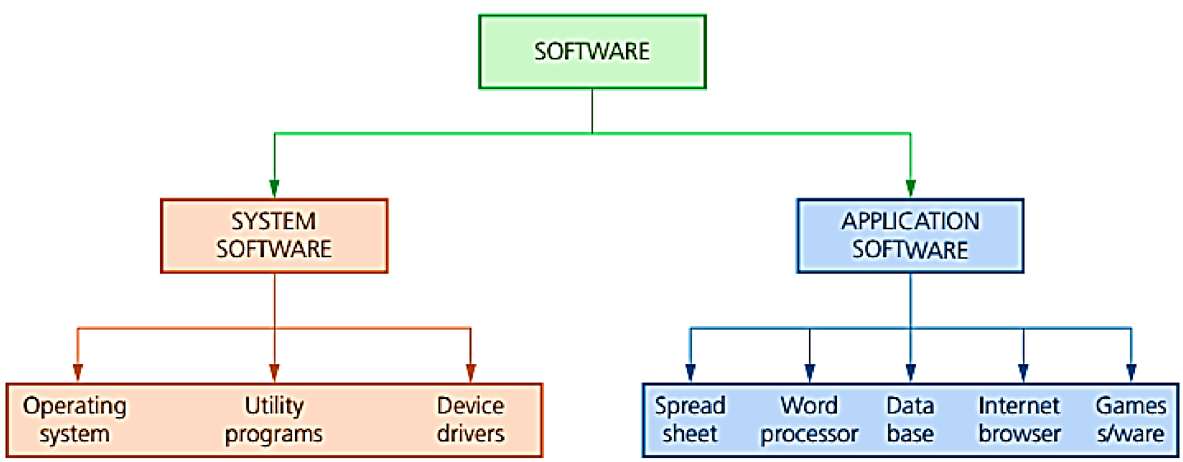
Chapter 4 : Software

System Software & Application Software



Software and hardware hierarchy



Software Hierarchy

General Features Of System Software	General Features of Application Software
» Set of programs to control and manage the operation of computer hardware. » Provides a platform on which other software can run. » Required to allow hardware and software to run without problems. » Provides a human computer interface (HCI). » Controls the allocation and usage of hardware resources.	» Used to perform various applications (apps) on a computer. » Allows a user to perform specific tasks using the computer's resources. » May be a single program (for example,notepad) or a suite of programs (for example, Microsoft Office). » User can execute the software as and when they require.

APPLICATION SOFTWARE

These are programs that allow the user to do specific tasks. Examples include:

WORD PROCESSOR

Word processing software is used to manipulate a text document, such as an essay or a report. This software provides tools for copying, deleting and various types of formatting. Some of the functions of word processing software include:

- creating, editing, saving and manipulating text
- copy and paste functions
- spell checkers and thesaurus
- import photos/images into a structured page format
- translation into a foreign language

SPREADSHEET

Spreadsheet software is used to organize and manipulate numerical data (in the form of integer, real, date, and so on).Numbers are organized on a grid of lettered columns and numbered rows.

Some of the functions of spreadsheets include:

- use of formulas to carry out calculations
- ability to produce graphs
- ability to do modelling and "what if" calculations.

DATABASE

Database software is used to organise,manipulate and analyse data. A typical database is made up of one or more tables. Tables consist of rows and columns. Each row is called a 'record' and each column is called a 'field.'

Some of the functions include:
• ability to carry out queries on database data and produce a report
• add, delete and modify data in a table

CONTROL AND MEASURING SOFTWARE

Control and measuring software is designed to allow a computer or microprocessor to interface with sensors so that it is possible to:
• measure physical quantities in the real world (such as temperatures)
• to control applications (such as a chemical process) by comparing sensor data with stored data and sending out signals to alter process parameters (e.g. open a valve to add acid and change the pH)

APPS

Apps is short for applications – a type of software. They normally refer to software which runs on mobile phones or tablets.

Common examples of apps include:
• video and music streaming
• GPS (global positioning systems – help you find your way to a chosen location)
• camera facility (taking photos and storing/manipulating the images taken).

PHOTO EDITING SOFTWARE

Photo editing software allows a user to manipulate digital photographs stored on a computer; for example, change brightness, change contrast, alter colour saturation or remove "red eye". They also allow for very complex manipulation of photos (e.g. change the features of a face,combine photos, alter the images to give interesting effects and so on). They allow a photographer to remove unwanted items and generally "touch up" a photo to make it as perfect as possible.

VIDEO EDITING SOFTWARE

Video editing software is the ability to manipulate videos to produce a new video.It enables the addition of titles, colour correction and altering/adding sound to the original video.

Essentially it includes:
• rearranging, adding and/or removing sections of video clips and/or audio clips
• applying colour correction, filters and other video enhancements
• creating transitions between clips in the video footage

GRAPHICS MANIPULATION SOFTWARE

Graphics manipulation software allows bitmap and vector images to be changed.Bitmap images are made up of pixels which contain information about image brightness and colour. Bitmap graphics editors can change the pixels to produce a different image. Vector graphic editors operate in a different way and don't use pixels – instead they manipulate lines,curves and text to alter the stored image as required. Both types of editing software might be chosen depending on the format of the original image.

SYSTEM SOFTWARE

These are programs that allow the hardware to run properly and allow the user to communicate with the computer examples include:

COMPILERS:

A compiler is a computer program that translates a program written in a high-level language (HLL) into machine code (code which is understood by the computer) so that it can be directly used by a computer to perform a required task. The original program is called the source code and the code after compilation is called the object code. Once a program is compiled, the machine code can be used again and again to perform the same task without re-compilation.

Examples of high-level languages include: Java, Python,Visual Basic, Fortran, C++ and Algol.

LINKERS:

A linker (or link editor) is a computer program that takes one or more object file produced by a compiler and combines them into a single program which can be run on a computer.

For example, many programming languages allow programmers to write different pieces of code, called modules, separately. This simplifies the programming task since it allows the program to be broken up into small, more manageable sub-tasks.However, at some point, it will be necessary to put all the modules together to form the final program. This is the job of the linker.

DEVICE DRIVERS

A device driver is the name given to software that enables one or more hardware devices to communicate with the computer's operating system. Without drivers, a hardware device (for example,a computer printer) would be unable to work with the computer. All hardware devices connected to a computer have associated drivers. As soon as a device is plugged into the USB port of a computer,the operating system looks for the appropriate driver. An error message will be produced if it can't be found.

Examples of drivers include: printers, memory sticks,mouse, CD drivers, and so on.

OPERATING SYSTEMS (O/S)

The operating system (OS) is essentially software running in the background of a computer system. It manages many of the basic functions. Without the OS, most computers would be very user-unfriendly and the majority of users would find it almost impossible to work with computers on a day-to-day basis.

For example, operating systems allow:

- input/output operations
- users to communicate with the computer (e.g. Windows)
- error handling to take place
- the loading and running of programs to occur
- managing of security (e.g. user accounts, log on passwords).

UTILITIES

Utility programs are software that are designed to carry out specific tasks on a computer. Essentially, they are programs that help to manage, maintain and control computer resources.

Examples include:

- anti-virus (virus checkers)
- anti-spyware
- back-up of files
- disk repair and analysis
- file management and compression
- security
- screensavers
- disk defragmenter/ defragmentation software.